Cultural Analytics of Gender Representation in Film: A James Bond Case Study

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1. Introduction

Media has a long history of portraying gender in their stereotypical gender roles. Especially older blockbuster films such as Indiana Jones [1] portray the man (Dr. Jones) as fearless "tough guy" who gets what he wants. Women on the other hand are portrayed as love interest, hysterical, and afraid. This portrayal seems out of date, and films today, albeit still not perfect in their gender representation, would be unlikely to release with such messages. But when did this change occur? Was the change gradual, or did some events/movements bring forward bigger leaps of change?

In this document we investigate how the portrayal of women has changed in the last decades. Specifically, we have chosen the James Bond film series as subject of research. To make our findings quantifiable, the films were analyzed using pretrained AI models, notably YOLO [2] and CLIP [3]. Given the use of pretrained models, we also investigate the validity of our method.

In the process we have created an analysis framework, which can be used for other research questions. This resulting framework is corpus agnostic, and can be adapted to detect different personality traits. It is open source and free to use [4].

2. Research Question

In this document we question whether women in James Bond are depicted in stereotypical gender roles, and whether this depiction has changed over the years. Based on our observation from Indiana Jones, we have chosen to measure two traits: Sexualization and Agency (describing whether a person is in control of the situation, or whether the person is helpless).

3. Related Works

Neuendorf (2009) [5] analyzed the portrayal of women in the first 20 James Bond films. A trend was discovered that women in later movies were subjected to more violence and were involved in more sexual activities. However the authors also pointed out that the roles of women changed from very limited roles to more autonomous and active roles. Our study differs from Neuendorf's in that we are observing the appearance rather than the activity of the different roles.

4. Dataset

The corpus chosen for our analysis is the James Bond 50th Anniversary DVD Collection. It was chosen for several reasons: **Long Running Series:** The collection contains films from 1962 to 2008 allowing for an observation period of 46 years. **Series Containing Many Films:** The Collection contains 22 films. The release interval varies from film to film, but overall they were released mostly regularly. The longest time between releases was 6 years, while the average is around 2 years. **Limiting Confounding Factors:** By limiting our observation to one series, we limit confounding factors. James Bond films all are of the same genre, mostly consist of the same slowly changing cast and crew, and follow similar plots. This makes comparison easier than comparing films of completely different genres (such as Indiana Jones, and Barbie). **Notoriety For Gender Representation:** The James Bond series is notorious for its representation of women, even coining the term Bond girl [6]. The prior research on this Series allows us to validate our methodology, to test whether it can be applied to other corpora. The use of only James Bond movies however also sets limitations for our study, as discussed later.

The films were digitized using VLC. Due to copyright restrictions, they are not provided with the code-base. The code-base however is corpus agnostic and can be applied to any other collection of films.

5. Methods

To retrieve the portrayal of women from the raw film data several steps were required. First, the individual films were split into their scenes and saved as images. Then, object detection was performed to get bounding boxes of persons in the scene images. Finally, trait and gender detection was performed on the detected bounding boxes.

5.1. Scene detection

First, the video data was split into scene data. We used PySceneDetect [7] to automate this process. For every scene detected by PySceneDetect, one frame in the middle of the scene was saved as image. This resulted in 800-2000 frames per film totaling 31000 frames.

5.2. Person detection

The next step applied person detection on every scene frame. This was performed with YOLO [2]. The pretrained yolo11 model was used, limited to only person detection. Furthermore the minimum confidence was set to 0.5. This was used in order to limit person detection to people in the foreground.

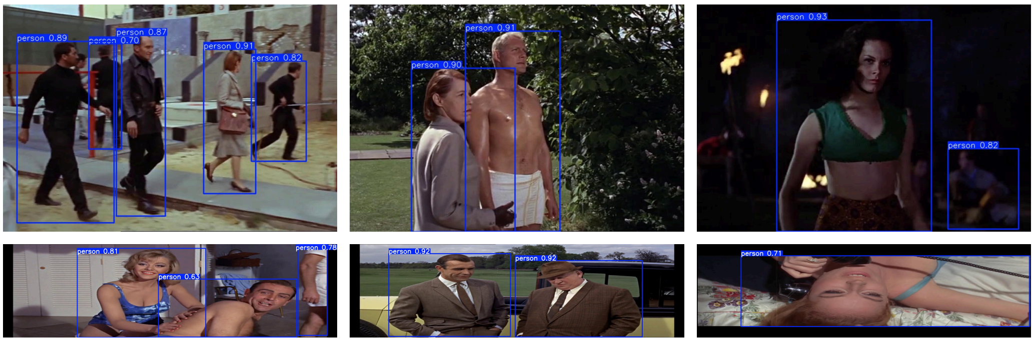


Figure 1: persons detected with YOLO11

5.3. Trait and Gender Detection

The last step in the pipeline performs trait and gender detection for each person bounding box, which YOLO detected. Trait and Gender detection are both performed using CLIP [3]. **Images** used are cropped along the bounding boxes to only contain the detected person and the images were processed in a batch size of 100 images. **Text** prompts were used to determine gender, and prompt ensembles were used for traits to improve performance [8].

trait	**stereotypical female traits**	**stereotypical male traits**
gender	"a photo of a woman"	"a photo of a man"
agency	"a person overwhelmed with the situation" "a person who is passive" "a submissive person"	"a person in control of the situation" "a person who gets what they want" "a dominant person"
sexualization	"a sexualized person" "a person in a swimsuit" "a scantily clad person"	"a professional person" "a person in formal attire" "an adequately dressed person"

The CLIP similarity values are averaged within the ensemble and then tranformed into scores from 0 to 1. 0 indicates stereotypical male traits (dominant and professional), and 1 indicates stereotypical female traits (sexualized and submissive).

6. Limitations

Our choice of corpus and methods comes with several limitations:

Corpus: While limiting our observation to the James Bond film series comes with many benefits, there are also some limitations associated with it. While the 22 films provide a large sampel size for a film series, it is still prone to film-to-film variations. Limiting ourselves to James Bond also limits the ability to generalize our findings to other films.

Method: Using pretrained models also brings a set of limitations. Object detection recognizes partial persons, where analyzing traits is not helpfull (see Fig. 3). Trait Analysis with CLIP does not differentiate between the person of interest, and background (see Fig. 5). The used models themselves can be biased (i.e. models attributing higher sexualization scores for women, regardless of input). And finally the prompts use abstract concepts such as “sexualized” or “dominant”, which might be difficult for the models to understand. We hope to mitigate this by using prompt ensembles.

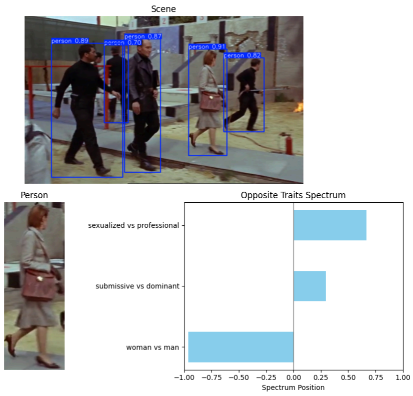


Figure 2: The model correctly recognizes five persons in the image. It recognizes the woman and interprets her as dominant and professional. This aligns with the film, as the woman is the leader of the depicted group

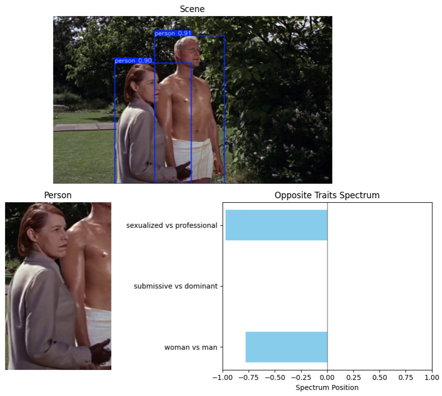


Figure 3: The gender is correctly inferred, but the degree of sexualization is clearly wrong. It seems the naked man in the background contributes to this score, even though he is not the person of interest

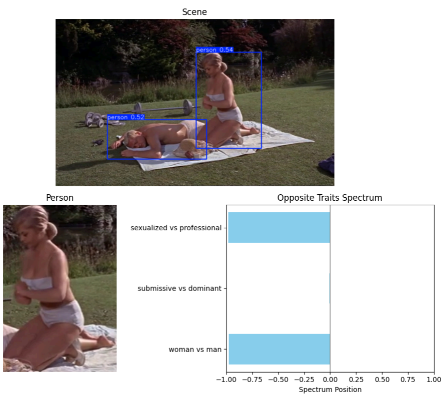


Figure 4: The model recognizes to persons in the image. It recognizes the woman as woman and that her depiction is sexualized

Figure 5: While the gender is correctly inferred, only the arm is visible. The model still predicts the degree of sexualization and dominance.

7. Findings

We visualized the agency score by gender and film and fitted a linear regression model to the data (see Fig. 7). We found that, in contrast to the representation of women in Indiana Jones, the models predicted that women in James Bond were not significantly more helpless than men. This fits the subjective observation we have made, that women in James Bond are commonly portrayed as confident and seductive.

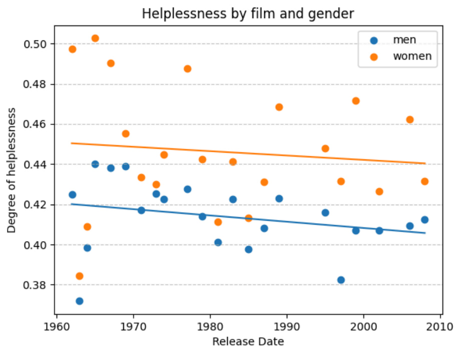


Figure 7: degree of helplessness in films by gender. Observed data and fitted linear regression model

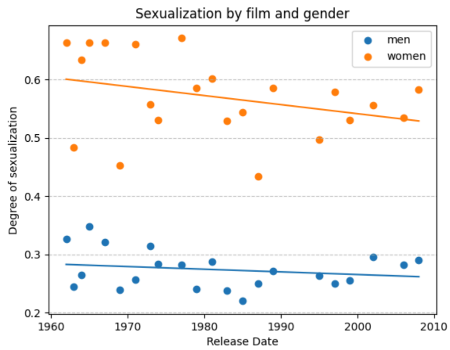
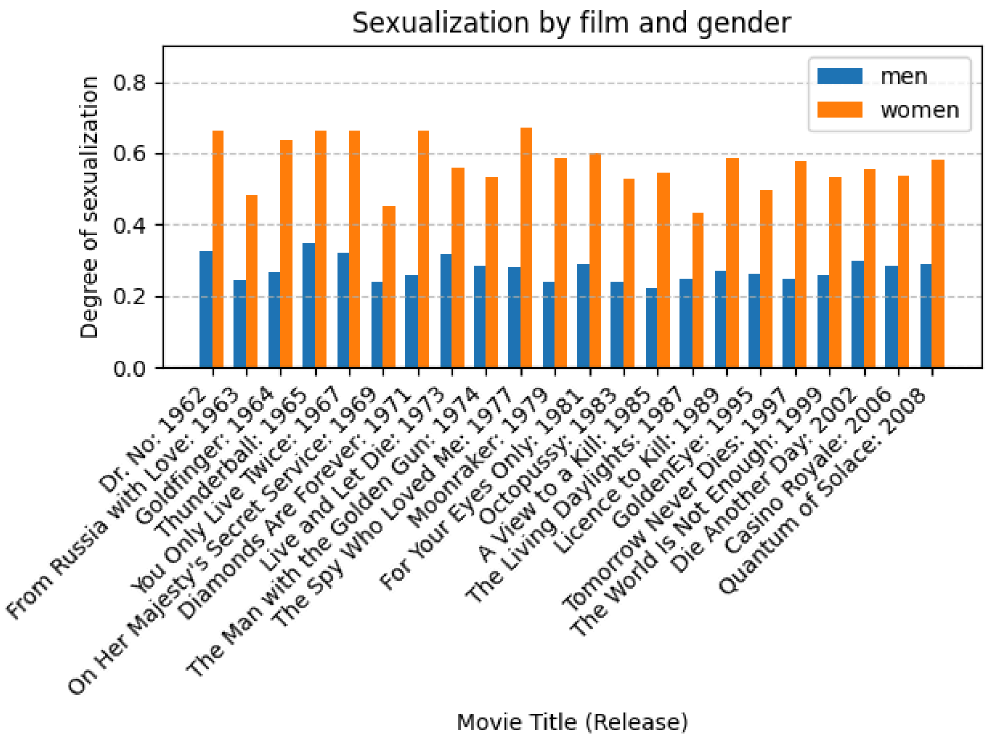
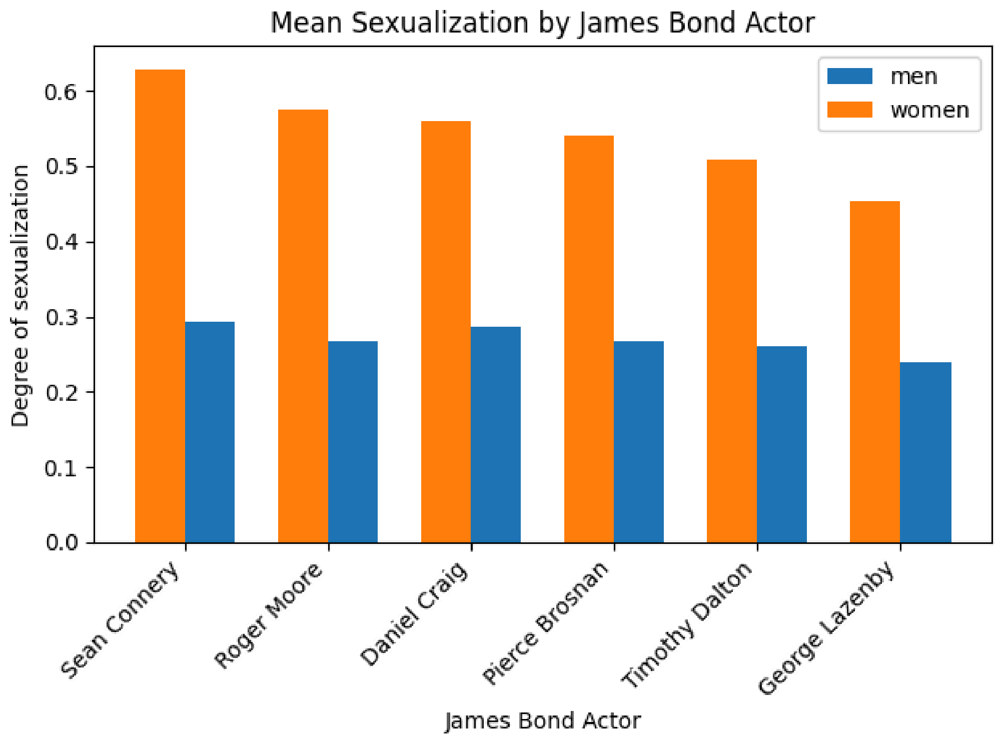


Figure 8: degree of sexualization in films by gender. Observed data and fitted linear regression model

For the sexualized representation however, there was a stark difference between gender (see Fig. 8). The mean sexualization score in men is 0.27, whereas in women it is 0.56. The Linear regression model shows a trend toward less sexualization both in men and women. Where the sexualization scores for men and women in 1962 were at 0.28 and 0.60, in 2008 those scores dropped to 0.26 and 0.53.



To validated our methods we plotted sexualization scores by titles where some outliers became apparent, notably *From Russia with Love: 1963* and *On Her Majesty's Secret Service: 1969*. Those outliers can be well explained by the film plots. In *From Russia with Love* James Bond is accompanied on his mission by a female spy whereas in other films most women are only acquaintances he sleeps with. Thus, there is more screen-time to normalize the sexualization score of women. In *On Her Majesty's Secret Service* James Bond marries a woman. The film is thus more romantic and less sexualized than others in the series



To further validate our methods we investigated the mean sexualization score by James Bond Actor [9]. Daniel Craig is deemed to be the least sexualized James Bond authors, however with our methods this could not be validated. The corpus used for this observation however only contained two films (*Casino Royale* and *Quantum of Solace*), missing *Skyfall*, *Spectre*, and *No Time to Die*. Including those missing titles into the analysis might yield different results.

8. Conclusion

In conclusion, we found that there is a slight trend towards less sexualization in the James Bond film Series. This trend exists both in men and women, but is weaker in men. We could not observe sudden changes, which could have been expected during waves of feminism for instance [10]. Instead the change was gradual and slow.

Outliers helped in validating our methods, however the strongly varying score also highlights, that the limited sample size of 22 is not sufficient to generalize from the portrayal of women in James Bond to the overall portrayal of women in film. Further studies with larger corpora are warranted in order to monitor general trends. Other long-running film series would be well suited to limit confounding factors.

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